Practical No 10.

Aim: Explore Agile Methodology Tool Jeera

Agile methodology is an iterative and incremental approach to software development. Instead of developing the entire software at once, the project is divided into smaller units of work called iterations or sprints.

An Agile tool helps the development team to:

- Create and manage requirements.
- Maintain a product backlog.
- Plan sprints.
- Create and assign tasks.
- Track task progress.
- Monitor team performance.
- Manage bugs and issues.
- Review completed work.
- Maintain project history.

Examples of Agile Tools

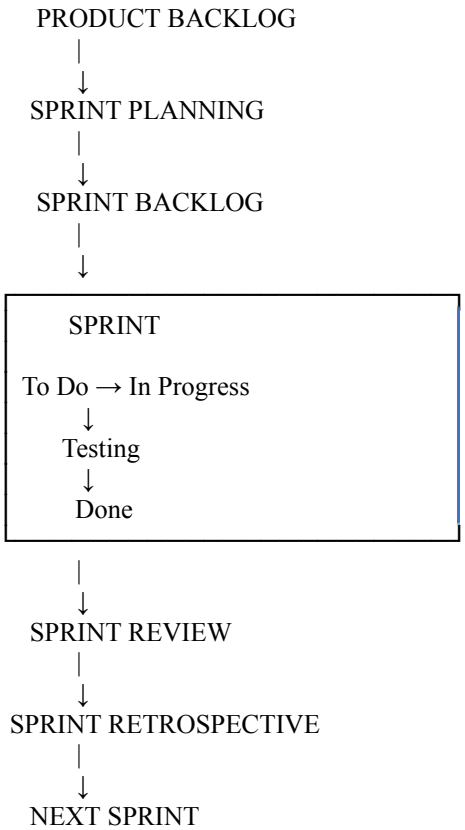
Tool	Main Use
Jira	Scrum, Kanban, issue and sprint management
Trello	Kanban-style task management

For this practical, **Jira can** be considered as the Agile project management tool.

1. Jira -Agile Scrum Framework

Jira can be used to implement the **Scrum framework**.

The basic Scrum workflow is:



- **Important Components of an Agile Tool**

1 Project
A project is a collection of all activities related to a particular software system.
Example: Library Management System

2 Product Backlog
The Product Backlog contains all requirements, features, improvements, and tasks that need to be developed.

For example:

ID	Backlog Item	Priority
US01	Student can search books	High
US02	Librarian can issue books	High
US03	Librarian can return books	High
US04	System calculates fine	Medium
US05	Admin can generate reports	Medium
US06	Student can reserve books	Low

- **Creating a Sprint**

A Sprint is a fixed development period during which selected backlog items are completed.

For example:
Sprint 1
Duration: 2 weeks

Task	Status
Design login	Done
Member registration	Done
Add books	In Progress
Search books	To Do
Issue books	To Do

After Sprint 1, completed work is reviewed and remaining work can be moved to a subsequent sprint.

- **Scrum Board**

The Scrum board visually displays the status of work.

TO DO	IN PROGRESS	TESTING	DONE
Search Book Fine Module	Issue Book Return Book	Login Member Mgmt	Register Add Book

When development starts, a task moves from **To Do → In Progress**.
After development, it moves to **Testing** and finally to **Done**.

- **Assigning Tasks**

The Agile tool allows tasks to be assigned to team members.
Example:

Task	Assigned To	Status
Login Module	Developer 1	Done
Book Module	Developer 2	In Progress
Fine Module	Developer 3	To Do
Testing	Tester	In Progress

This helps the Scrum team understand who is responsible for each task.

- **Managing Bugs**

An Agile tool can also be used for bug tracking.
Example
Bug ID: BUG-01
Description: Fine is calculated incorrectly for overdue books.
Priority: High

Status: Open
Open → In Progress → Testing → Resolved → Closed
The developer fixes the bug, the tester verifies the correction, and the issue is then closed.

- **Sprint Review**

At the end of a sprint, the team demonstrates the completed functionality to the relevant stakeholders.
For the Library Management System, the team may demonstrate:

- Member registration
- Book search
- Book issue
- Book return
- Fine calculation

The feedback can be converted into new or modified backlog items.

- **Sprint Retrospective**

The retrospective is conducted after a sprint to identify:

- What went well?
- What problems occurred?
- What can be improved?
- What actions should be taken in the next sprint?

Example:

Observation	Improvement
Testing started late	Start testing earlier
Requirements were unclear	Clarify user stories
Tasks were too large	Divide tasks into smaller items

- **Agile Workflow for Library Management System**

The complete workflow can be represented as:

Identify Requirements
↓
Create Product Backlog
↓
Prioritize User Stories
↓
Sprint Planning
↓
Create Sprint Backlog
↓
Assign Tasks
↓
Development
↓
Testing
↓
Sprint Review
↓
Retrospective
↓
Next Sprint

- **Example Sprint Plan**

Sprint	Features
Sprint 1	Login, Member Registration, Book Management
Sprint 2	Book Search, Book Issue, Book Return

Sprint	Features
Sprint 3	Fine Management, Reports
Sprint 4	Reservation, Testing and Improvements

- **Advantages of an Agile Tool**
  1. Provides centralized project management.
  2. Helps maintain the product backlog.
  3. Supports sprint planning.
  4. Makes task assignment easier.
  5. Provides visual task tracking.
  6. Helps manage bugs and issues.
  7. Provides project progress reports.
  8. Maintains a history of changes.
  9. Improves communication among team members.
  10. Supports continuous improvement.

- **Result:** The Agile methodology tool was successfully explored. The tool was studied for creating a project, preparing the product backlog, creating user stories, planning sprints, assigning tasks, tracking work through a Scrum board, managing bugs, and monitoring project progress using Agile practices.